

Certified analytical bounds for hypersonic reachable sets

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Abstract

Computing exact reachable sets for six-degree-of-freedom hypersonic entry vehicles is intractable for grid-based dynamic programming and compromised by the floating-point approximations of numerical solvers. This research introduces a purely analytical framework to compute mathematically certified bounds using exact interval arithmetic and occupation measures. By modeling transcendental aerodynamics natively as interval differential inclusions, the method circumvents artificial state augmentation and numerical optimization entirely. Utilizing a thin-atmosphere limit profile decomposition to separate the equilibrium-glide skeleton from compact skip events, the framework yields explicit, closed-form algebraic bounds that provide a strict mathematical certificate in the minimal physical state space.

Open items

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The problem

The reachable set of an entry vehicle (every state it can be driven to under admissible control) is what safety and threat analyses need bounded. Computing it exactly means solving a Hamilton–Jacobi–Isaacs equation on a grid, which is exponential in state dimension and capped in practice near four or five states [1, 2]. A rigid-body entry vehicle carries thirteen states, $x = (r, \theta, \phi, u, v, w, q_0, q_3, \omega_{1:3}) \in \mathbb{R}^{13}$, and twenty-five once ablation recession, integrated heat load and aerothermoelastic deformation are coupled [3].

So the exact grid-HJI method quits before the real problem starts.

The standard polynomial barrier with coefficients obtained from an interior-point solver has analytical *form* but inherits the solver’s accuracy [4], so it is not a certificate; a numerical solver’s failure modes are documented, and they include the loss of strict feasibility on which every rounding-based scheme depends [5, 6]. There are then two problems to solve, not one: dimension, and the provenance of the guarantee. The second is not a matter of care. Interval arithmetic is the setting in which a guarantee has meant, for forty years, an enclosure rather than an approximation, and the property that makes an enclosure valid at all—inclusion isotonicity—is exactly what a reimplementations loses first; not every law of real arithmetic survives the move to intervals, which is the original form of the wrapping effect [7, 8]. Applied interval analysis supplies the machinery the pipeline would be built from, and it is deliberately conservative: certified set inversion, deterministic intersection geometries, and a customized exact-arithmetic integrator to ensure floating-point inaccuracies do not compromise the final bounds [9].

Exact-Arithmetic Computational Pipeline The proposal demands a guarantee verified in exact arithmetic rather than produced by a numerical solver. Floating-point inaccuracies inherent to standard simulation tools compromise the analytical certificate. Must implement a custom, exact-arithmetic interval integrator (likely leveraging C/C++ or specialized rational arithmetic libraries). This pipeline must process the hybrid dynamics, manage the boundary-layer bubble transitions, and compute the intersection geometries deterministically to verify the final bounds.

The approach

Embed the hybrid flight dynamics into a Banach space of occupation measures. The nonlinear flight mechanics $\dot{x} = f_q(x, c, t)$ become *linear*: the Liouville equation is a linear constraint on a measure μ over $[0, T] \times \mathcal{X} \times U$, and $\text{supp } \mu_T$ outer-approximates the reachable set. So far this is the occupation-measure program of Lasserre, Henrion and collaborators [10, 11]. The embedding buys linearity at the cost of uniqueness, and the way it is recovered is standard: a trajectory family satisfying the Liouville constraint lies in the convex hull of the generalized curves, so measures correspond to convex mixtures of trajectories rather than to one of them [12, 13]. The same relaxation is what admits the interval envelope, and that is the first thing that has to be built rather than borrowed. Because the true aerodynamics are transcendental, they are enclosed in strict intervals, and a vector field selected from an interval is a set-valued map; the resulting differential inclusion is the object the measure must describe, not the nominal field [14, 15]. Occupation-measure theory for inclusions of exactly this kind already exists and must be cited as prior art: it defines an occupational invariant measure without assuming solution uniqueness, which is the same move made here [16]. What it assumes is worth stating plainly, because it is where the entry problem is likely to fail: the vector fields must satisfy a linear growth condition, and the singular-perturbation results additionally require a uniform bound on the fast state along the whole ε -family [16, 17].

Liouville Equation for Differential Inclusions The current framework embeds the flight mechanics $\dot{x} = f_q(x, c, t)$ as a linear Liouville constraint on an occupation measure μ over $[0, T] \times \mathcal{X} \times U$. By bounding true non-polynomial aerodynamics within strict intervals, the vector field becomes a differential inclusion $\dot{x} \in \mathcal{F}(x, c, t)$. We must formulate a robust Liouville equation capable of acting on set-valued maps, and rigorously prove that the support of the resulting occupation measure, $\text{supp } \mu_T$, securely outer-approximates all possible trajectories generated by any vector field selected from the interval inclusion.

The new content is what happens in the thin-atmosphere limit $\varepsilon_{\text{atm}} = H_s/R_e \rightarrow 0$. Rescaled by the phugoid-to-secular time ratio (which, as in the classical Chapman–Loh variables, enters as $\sqrt{\varepsilon_{\text{atm}}}$, not ε_{atm} [18, 19]), the occupation-measure family *is not tight*. Mass concentrates onto the low-dimensional equilibrium-glide slow manifold $\Lambda = L \cos \sigma / [m(g - V^2/r)] = 1$.

This is precisely Lions’ concentration alternative, read on the defect measure of the rescaled family [20]. The bubbles are concentration at a shrinking timescale, so the operative half of the principle is the dilation case rather than the translation case: it is the limit-case modification that applies [21]. Rather than restore compactness, we extract the profile decomposition the dichotomy hands us:

- a finite-dimensional *skeleton* $\mathcal{S}$ —the concentration component, the glide manifold as a graph over its retained coordinates, for which a rigorous characterization with quantitative accuracy estimates already exists [22];
- one *boundary-layer bubble* per skip event, the analogue of a concentrating profile: a compact impulse set $\mathcal{I}_j$ of dimension $\leq m$, carried by an internal scaling group at each atmospheric pass, with orthogonality read off the scale-and-core separation condition that governs concentrating profiles generally [23];

Wrapping Effect Suppression in Boundary-Layer Bubbles Boundary-layer bubbles represent discrete atmospheric skip events as compact impulse sets $\mathcal{I}_j$ carried by internal scaling groups. Because our framework fixes time as the independent variable, integrating interval bounds through highly dynamic skip events will induce severe “wrapping effects” (artificial inflation of bounding boxes). Must develop advanced intersection geometries (zonotopes, Taylor models, etc.) integrated into the profile decomposition to prove geometries maintain tight corridor bounds through multiple skip events without the reachable set exploding prematurely.

- a *compact residual* with a quantitative bound, $O(\sqrt{\varepsilon})$ in the corridor metric, in a norm chosen so that the loss of compactness actually disappears rather than merely being hidden [24].

The number of bubbles is bounded *a priori*, $K \leq K_{\max}(Q_{\max})$, by the integrated heat-load budget (a trajectory can only afford so many skips [25, 26]). The wrapping effect to be suppressed here is not new: for linear systems it is already quantified, and the term is first order in the time step and exponential in the horizon times the matrix norm [27]. The multi-parameter separation of the three scaling groups (atmospheric, attitude, elastic) is the Bahouri–Gérard almost-orthogonality argument, transported to defect measures; it is the one genuinely new obligation and it sits on the critical path.

The argument worth transporting is the constructive one: splitting a bounded sequence into an oscillatory and a singular part at each scale forces those scales to be orthogonal, and iterating extracts a pairwise orthogonal family, so orthogonality is produced by the construction rather than assumed by it [28].

Bahouri-Gérard Under Interval Disturbances The framework relies on the Bahouri-Gérard almost-orthogonality argument to separate the multi-parameter scaling groups (atmospheric, attitude, and elastic). Under interval aerodynamics, these groups couple with an adversarial, bounded uncertainty. We must prove the non-interaction theorem for the defect measures while accounting for interval bounds, guaranteeing that the gap between the upper and lower aerodynamic bounds does not cause mass to leak between scaling groups during atmospheric transition, preserving the compact residual bound $O(\sqrt{\epsilon})$.

The target statement is

$$\mathcal{R}_{\text{cert}}(w) \subseteq \underbrace{\left(\Phi_{\mathcal{S}}(\Theta) \right)}_{\text{skeleton}} \oplus \underbrace{\left(\bigoplus_{j=1}^K \mathcal{I}_j \right)}_{\text{bubbles}} \oplus \underbrace{\mathbb{B}(C_{\mathcal{R}}\sqrt{\epsilon})}_{\text{residual}},$$

uniformly over the admissible family generated by the uncertainty set, not along a nominal trajectory. The exact residual corridor constant $C_{\mathcal{R}}$ and the maximum bubble count $K_{\max}$ are derived as explicit, closed-form algebraic expressions directly from the physical corridor bounds and aerodynamic interval widths, completely independent of numerical optimization logic. Requiring the constants in closed form, rather than as the output of a solver, is what separates a certificate from a computation: exact rational certificate representations are known to exist, with a documented bit-complexity that grows with the number of variables [29, 30]. Because interval arithmetic evaluates transcendentals natively, the traditional algebraic adjunctions required to force the system into a polynomial format [31] are obsolete. The augmented state space is collapsed back to its minimal physical dimensions, retaining only the adjunctions genuinely necessary to avoid coordinate singularities.

Analytical Derivation of Corridor Constants The ultimate guarantee relies on deriving the residual corridor constant $C_{\mathcal{R}}$ and the maximum boundary-layer bubble count $K_{\max}(Q_{\max})$; The number of bubbles is bounded a priori by the integrated heat-load budget. We must then derive explicit, closed-form algebraic expressions for $C_{\mathcal{R}}$ and $K_{\max}$ strictly from the physical corridor bounds, initial conditions, and the interval widths of the aerodynamics, entirely without relying on numerical interior-point solvers.

State Space Minimization and Kinematic Trade-offs The shift to exact interval arithmetic renders traditional Savageau-style polynomial adjunctions (e.g., $y = \sqrt{\rho/\rho_0}$, $\rho = 1/||\vec{r}||$) completely obsolete, as transcendentals are evaluated natively. We must formally strip away these algebraic embeddings and collapse the theoretical state space back to its minimal physical dimensions, which inherently tightens bounds by reducing susceptibility to the wrapping effect. The remaining open implementation task is determining if retaining purely kinematic adjunctions (such as quaternions to avoid Euler angle singularities) is computationally necessary to prevent interval bounding boxes from artificially inflating, or if the raw 13-state formulation is optimal for the pipeline’s intersection geometries.

A Coupled Entry Dynamics and Polynomial Embedding

A.1 Coupled 6-DOF Entry Dynamics

The entry vehicle state comprises the coupled translational, rotational, thermal, ablative, and modal subsystems. Position is carried in Cartesian Earth-centred Earth-fixed (ECEF) components and attitude as a quaternion from the ECEF frame to the body frame. No angle is a state, so the dynamics have no coordinate singularities anywhere on the globe; geocentric latitude and longitude are read out from the position after integration.

Frames.

- $\mathcal{E}$: Earth-centred Earth-fixed (ECEF), rotating with the constant planetary rate $\vec{\omega}_E = \omega_E \hat{z}_{\mathcal{E}}$.
- $\mathcal{B}$: body frame, fixed at the instantaneous center of mass.

The attitude quaternion $\vec{q} = (q_0, q_1, q_2, q_3)^\top$ carries $\mathcal{E} \rightarrow \mathcal{B}$, with direction-cosine matrix

$$\mathbf{R} \equiv \mathbf{R}_{B/E}(\vec{q}) = \begin{pmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1q_2 + q_0q_3) & 2(q_1q_3 - q_0q_2) \\ 2(q_1q_2 - q_0q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2q_3 + q_0q_1) \\ 2(q_1q_3 + q_0q_2) & 2(q_2q_3 - q_0q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{pmatrix}, \quad (\text{A.1})$$

quadratic in $\vec{q}$.

Position Kinematics. With $\vec{r} = (r_1, r_2, r_3)^\top$ the ECEF position and $\vec{V} = (u, v, w)^\top$ the planet-relative velocity resolved in $\mathcal{B}$, the kinematic equation is:

$$\dot{\vec{r}} = \mathbf{R}^\top \vec{V}. \quad (\text{A.2})$$

Translational Dynamics. Newton's law holds in the inertial frame; transporting the planet-relative velocity through $\mathcal{E}$ into $\mathcal{B}$, with inertial body rates $\vec{\omega} = (p, q, r)^\top$, mass m , aerodynamic force $\vec{F}_a$, thrust $\vec{F}_t$, and the planetary rate resolved in the body frame $\vec{\omega}_E^{\mathcal{B}} = \mathbf{R} \omega_E \hat{z}_{\mathcal{E}}$, the translational dynamical equations are:

$$\dot{\vec{V}}|_{\mathcal{B}} = \frac{1}{m}(\vec{F}_a + \vec{F}_t) + \mathbf{R} \vec{g}(\vec{r}) - (\vec{\omega} + \vec{\omega}_E^{\mathcal{B}}) \times \vec{V} - \vec{\omega}_E^{\mathcal{B}} \times (\vec{\omega}_E^{\mathcal{B}} \times \mathbf{R} \vec{r}). \quad (\text{A.3})$$

Gravity, including the J_2 oblateness term, in ECEF components:

$$\vec{g}(\vec{r}) = -\frac{\mu}{\|\vec{r}\|^3} \left[\left(1 + \frac{3}{2} J_2 \frac{R_e^2}{\|\vec{r}\|^2} \left(1 - 5 \frac{r_3^2}{\|\vec{r}\|^2} \right) \right) \vec{r} + 3 J_2 \frac{R_e^2}{\|\vec{r}\|^2} r_3 \hat{z}_{\mathcal{E}} \right]. \quad (\text{A.4})$$

Remark A.1. This is the gradient of the J_2 potential $\frac{\mu}{\|\vec{r}\|} [1 - \frac{1}{2} J_2 (R_e / \|\vec{r}\|)^2 (3r_3^2 / \|\vec{r}\|^2 - 1)]$; its radial and northward components reproduce the spherical form, with $r_3 / \|\vec{r}\| = \sin \phi$.

Rotational Dynamics / Euler's Equations. With angular momentum $\vec{h} = \mathbf{I} \vec{\omega}$, combined aerodynamic / reaction control system (RCS) moments $\vec{M} = \vec{M}_a + \vec{M}_{\text{RCS}}$, and the inertia tensor $\mathbf{I} = \mathbf{I}(s, \eta)$ depending on recession depth s and modal amplitudes η (so that $\dot{\mathbf{I}} \vec{\omega}$ is the variable-inertia term), the rotational dynamical equations are:

$$\mathbf{I} \dot{\vec{\omega}} = \vec{M} - \vec{\omega} \times (\mathbf{I} \vec{\omega}) - \dot{\mathbf{I}} \vec{\omega}. \quad (\text{A.5})$$

Quaternion Kinematics. The quaternion is driven by the body rate *relative to* $\mathcal{E}$, $\vec{\omega}_{B/E} = \vec{\omega} - \vec{\omega}_E^{\mathcal{B}}$. For $\vec{a} = (a_1, a_2, a_3)^\top$,

$$\dot{\vec{q}} = \frac{1}{2} \Omega(\vec{\omega} - \mathbf{R} \omega_E \hat{z}_{\mathcal{E}}) \vec{q}, \quad \Omega(\vec{a}) = \begin{pmatrix} 0 & -a_1 & -a_2 & -a_3 \\ a_1 & 0 & a_3 & -a_2 \\ a_2 & -a_3 & 0 & a_1 \\ a_3 & a_2 & -a_1 & 0 \end{pmatrix}. \quad (\text{A.6})$$

Remark A.2. The rate is **polynomial** — bilinear in $(\vec{\omega}, \vec{q})$ and cubic in $\vec{q}$ through the Earth-rate term. With this Ω and $\mathbf{R}$ of Eq. (A.1), $\dot{\mathbf{R}} = -[\vec{\omega}_{B/E} \times] \mathbf{R}$ holds identically, and $\Omega(\vec{a})$ is skew for every $\vec{a}$, so $|\vec{q}|$ is preserved.

Thermal. With the density-square-root embedding $y = \sqrt{\rho/\rho_0}$ and $V = \sqrt{u^2 + v^2 + w^2}$, the modulus lift $\dot{\chi} = c \chi \dot{T}_w$ replaces T_w with a polynomial via Varma–Savageau–Voit recasting [31]. Stagnation-point heating (Sutton–Graves [32], with nose radius R_n , sea-level density ρ_0) equations are then given by:

$$\dot{Q}_{\text{tot}} = \dot{Q} = k_Q \sqrt{\rho_0/R_n} y V^3, \quad \dot{T}_w = \frac{\dot{Q} - (T_w - T_\infty)/\tau_{\text{cond}}}{C_w}. \quad (\text{A.7})$$

The correlation is used inside its stated validity range and not extrapolated past it: axisymmetric blunt bodies, gases in chemical equilibrium, enthalpy 2.3 to 116.2 MJ/kg, pressure 0.001 to 100 atm, and wall temperature at one of two tabulated values rather than a range[32, 33]. The chemical-equilibrium clause is the one most likely to fail, at the high-altitude end, which is where the nonequilibrium and rarefied treatments take over[34]. The density profile itself should come from a whole-atmosphere model rather than a spliced pair, since a splice would put a discontinuity in the vector field and defeat the polynomial recast[35].

Ablation. With recession rate coupled to heat flux and mass loss from char-layer blow-off, the ablative equations are given by:

$$\dot{s} = \alpha_s \dot{Q}, \quad \dot{m} = -\dot{m}_{\text{abl}}. \quad (\text{A.8})$$

Recession is a *state* with its own differential equation rather than a post-processed output because the interface condition requires the heat input to equal the heat conducted into the solid plus the heat absorbed in melting; the transformation that maps a receding boundary onto a fixed one is the classical one, and it costs only extra convective terms in products of states already carried[36]. A decoupled correlation for the recession rate is not obviously conservative: the surface energy balance couples to a multicomponent boundary layer with unequal diffusion coefficients, and the adequacy of a decoupled treatment depends on the material–environment pair[37].

Modal/Structural Elasticity. With piston-theory forcing and temperature-dependent natural frequency, per mode i our modal equations are:

$$\ddot{\eta}_i + 2\zeta_i \omega_i(T) \dot{\eta}_i + \omega_i(T)^2 \eta_i = Q_i(x, \eta, \dot{\eta}). \quad (\text{A.9})$$

Piston theory is a double expansion in thickness ratio and inverse Mach number valid when the contributions of order δ/M^3 and δ^2/M^2 are negligible, which is where its applicability conditions are stated and where they must be checked rather than assumed[38]. That choice is load-bearing

for the cost in SOS degree: the first-order point-function relation is affine in the normal component of fluid velocity, hence affine in the modal rates and free in the Gram table, whereas each further order costs one degree in the downwash and multiplies the modal block through. Carrying first order exactly and enveloping the higher-order residuals against the paper’s own smallness criterion is therefore the trade to state explicitly. The pressure band over which the aerodynamic law is the one being certified is likewise a checkable, state-dependent condition, and belongs in the constraint set rather than in an assumption[39].

A.2 Polynomial Embedding via Adjunction

A priori, the dynamics contain transcendental and rational terms. We repair each of these by adjoining one variable and one polynomial defining constraint, so the entire augmented system is polynomial in the augmented state:

1. **Transcendental Incidence:** $\alpha = \text{atan2}(w, u)$, $\beta = \arcsin(v/V)$ are inverse-trigonometric functions of body rates, we replace them with polynomial direction-cosine ratios $(u/V, v/V, w/V)$.
2. **Square-Root Speed:** Since $V = \sqrt{u^2 + v^2 + w^2}$ carries a square root, we adjoin V with the velocity magnitude constraint $V^2 = u^2 + v^2 + w^2$.
3. **Exponential Atmosphere:** Since $\rho = \rho_0 e^{-h/H_s}$ with $h = \|\vec{r}\| - R_e$ appears in $\bar{q} = \frac{1}{2}\rho V^2$, we adjoin $y = \sqrt{\rho/\rho_0}$ with rate $\dot{y} = -\frac{y}{2H_s} \frac{d}{dt}\|\vec{r}\|$, where the radial rate $\frac{d}{dt}\|\vec{r}\| = \varrho(\mathbf{R}\vec{r}) \cdot \vec{V}$ is polynomial once ϱ (item 4) is adjoined.
4. **Reciprocal Radius:** Gravity carries $\|\vec{r}\|^{-3}$, $\|\vec{r}\|^{-5}$, $\|\vec{r}\|^{-7}$, so we adjoin $\varrho = 1/\|\vec{r}\|$ with $\varrho^2 \|\vec{r}\|^2 = 1$, $\varrho > 0$, and rate $\dot{\varrho} = -\varrho^3(\mathbf{R}\vec{r}) \cdot \vec{V}$. Gravity is then polynomial:

$$\vec{g} = -\mu\varrho^3 \left[\left(1 + \frac{3}{2}J_2R_e^2\varrho^2(1 - 5r_3^2\varrho^2)\right)\vec{r} + 3J_2R_e^2\varrho^2 r_3 \hat{z}_{\mathcal{E}} \right]. \quad (\text{A.10})$$

No trigonometric adjunctions are needed: no angle is a state, and latitude dependence enters only through $r_3\varrho = \sin\phi$.

5. **Quaternion Norm:** $|\vec{q}|^2 = 1$ is already a polynomial invariant (skew-symmetry of Ω preserves it), no adjunction necessary.
6. **Heat / Load-Factor Square Roots:** $\dot{Q} \propto yV^3$ is already polynomial; however, $n = \|\vec{F}_a\|/(mg_0)$ requires squaring to $(\bar{q}S)^2(C_X^2 + C_Y^2 + C_Z^2) \leq (n_{\max}mg_0)^2$.

The Augmented State. Collecting all coordinates:

$$\tilde{x} = \left(\underbrace{r_1, r_2, r_3, u, v, w, q_0, q_1, q_2, q_3, p, q, r_b}_{13, \text{ core}}, \underbrace{y, V, \varrho, Q_{\text{tot}}, m, s, \chi}_{7, \text{ adjoined}}, \underbrace{\eta, \dot{\eta}}_{2n_{\eta}, \text{ modal}} \right). \quad (\text{A.11})$$

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